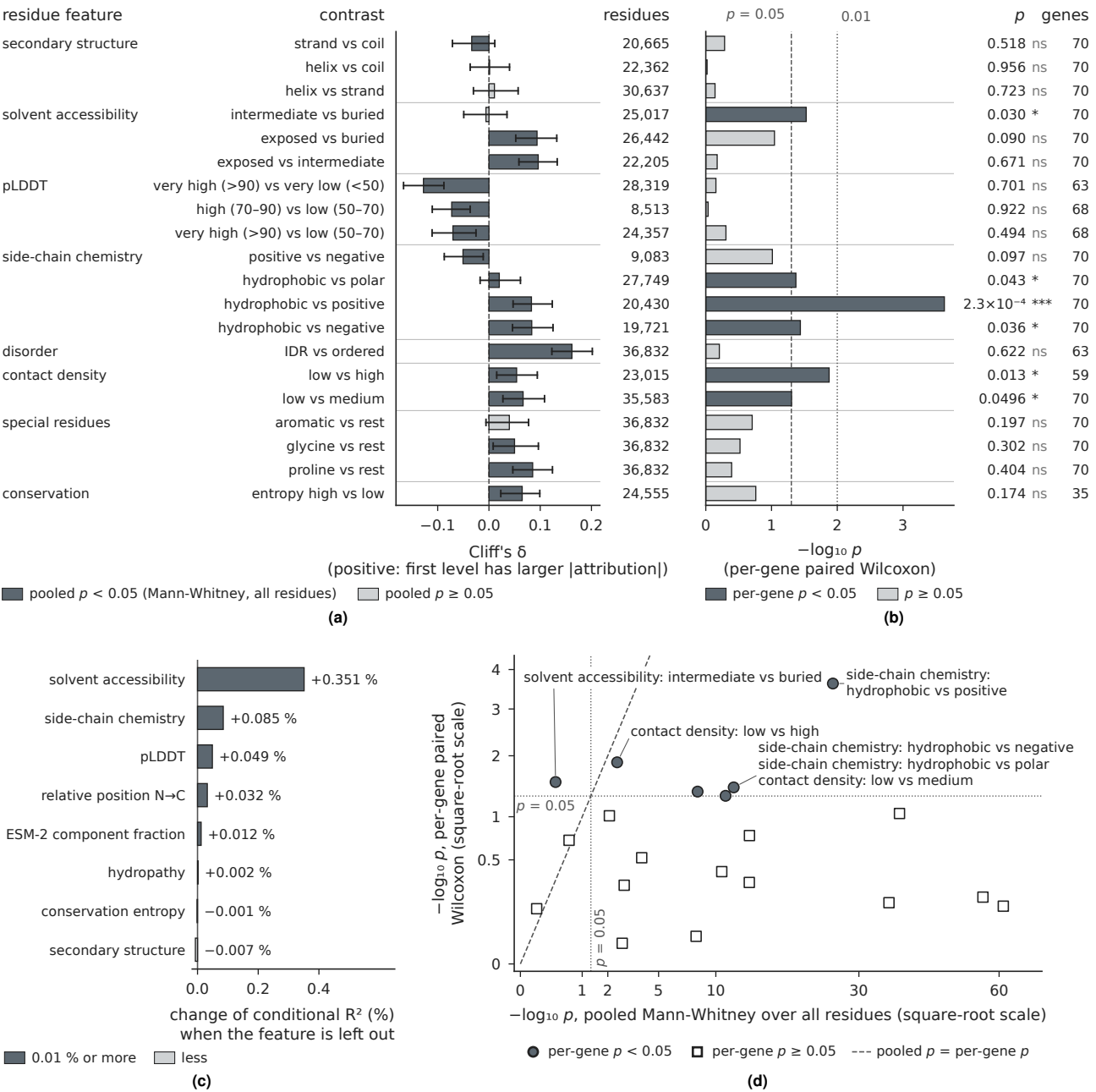

residue feature
contrast
residues
p = 0.05
0.01
p
genes
secondary structure
strand vs coil
20,665
0.518 ns 70
helix vs coil
22,362
0.956 ns 70
helix vs strand
30,637
0.723 ns 70
solvent accessibility
intermediate vs buried
25,017
0.030 * 70
exposed vs buried
26,442
0.090 ns 70
exposed vs intermediate
22,205
0.671 ns 70
pLDDT
very high (>90) vs very low (<50)
28,319
0.701 ns 63
high (70–90) vs low (50–70)
8,513
0.922 ns 68
very high (>90) vs low (50–70)
24,357
0.494 ns 68
side-chain chemistry
positive vs negative
9,083
0.097 ns 70
hydrophobic vs polar
27,749
0.043 * 70
hydrophobic vs positive
20,430
2.3×10⁻⁴ *** 70
hydrophobic vs negative
19,721
0.036 * 70
disorder
IDR vs ordered
36,832
0.622 ns 63
contact density
low vs high
23,015
0.013 * 59
low vs medium
35,583
0.0496 * 70
special residues
aromatic vs rest
36,832
0.197 ns 70
glycine vs rest
36,832
0.302 ns 70
proline vs rest
36,832
0.404 ns 70
conservation
entropy high vs low
24,555
0.174 ns 35
−0.1 0.0 0.1 0.2
Cliff's δ
(positive: first level has larger |attribution|)
0 1 2 3
−log₁₀ p
(per-gene paired Wilcoxon)
pooled p < 0.05 (Mann-Whitney, all residues)
pooled p ≥ 0.05
per-gene p < 0.05
p ≥ 0.05
(a)
(b)
solvent accessibility
+0.351 %
side-chain chemistry
+0.085 %
pLDDT
+0.049 %
relative position N→C
+0.032 %
ESM-2 component fraction
+0.012 %
hydropathy
+0.002 %
conservation entropy
−0.001 %
secondary structure
−0.007 %
0.0 0.2 0.4
change of conditional R² (%)
when the feature is left out
0.01 % or more
less
(c)
solvent accessibility: intermediate vs buried
side-chain chemistry: hydrophobic vs positive
contact density: low vs high
side-chain chemistry: hydrophobic vs negative
side-chain chemistry: hydrophobic vs polar
contact density: low vs medium
p = 0.05
p = 0.05
0 0.5 1 2 3 4
0 1 2 5 10 30 60
−log₁₀ p, per-gene paired Wilcoxon (square-root scale)
−log₁₀ p, pooled Mann-Whitney over all residues (square-root scale)
per-gene p < 0.05
per-gene p ≥ 0.05
pooled p = per-gene p
(d)